A Routing-Based Ensemble Framework for County-Level Power Outage Forecasting Under Extreme Weather

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Abstract

Short-term power outage prediction for Michigan's 83 counties is complicated by a deeply zero-inflated count distribution—approximately 70% of county-hours record zero outages—punctuated by rare, high-magnitude events driven by severe convective systems. Standard regression cannot simultaneously minimize error in quiet periods and capture catastrophic event dynamics, because optimizing a global loss function dominated by structural zeros inevitably under-weights the tail. This paper proposes a four-stage routing-based ensemble that decomposes the prediction problem by outage magnitude: a LightGBM classifier assigns each county-hour to one of four regimes identified through data-driven clustering; three diverse regressors provide within-regime predictions; Bayesian hyperparameter optimization adapts model parameters to each regime; and a meta-learner refines routing by exploiting out-of-fold prediction agreement across all base models. Prediction intervals are constructed via a magnitude-scaled adaptive approach whose parameters are tuned on validation residuals, yielding a Winkler score of 9.49 against 89.38 for a global residual quantile baseline—an 89% improvement. Gain-based feature importance aggregated across 336 trained models reveals a complete structural separation: the routing classifier relies on temporal autocorrelation to answer *when* extremes occur, while severity regressors rely exclusively on meteorological variables to answer *how severe*, with zero feature overlap. The dominant severity drivers are the Lifted Index, geopotential height at 400 mb, and near-surface specific humidity—a physically coherent finding that directly validates the two-stage architecture.

1 Introduction

Power outages impose substantial costs on households, businesses, and critical infrastructure. Accurate short-term forecasts enable utilities to pre-position repair crews, prioritize restoration of sensitive loads, and reduce the duration of interruptions. Yet unlike electricity price or demand forecasting, where targets exhibit smooth temporal structure, outage counts are non-negative integers dominated by structural zeros and governed by rare, physically catastrophic events that bear little resemblance to the quiet baseline.

Michigan's 83 counties are subject to severe convective weather associated with the Great Lakes, and the April–June 2023 training period is characterized almost entirely by a single severe thunderstorm on June 26, 2023. CAPE reached 14.6 times its seasonal background, precipitable water was elevated by 2.35×, and county-averaged hourly counts peaked at 1,040. Approximately 70% of county-hours record zero outages, with variance exceeding the mean by more than a factor of ten. A model optimizing mean squared error globally predicts near-zero for most samples, performing adequately on normal-case accuracy (s1) while failing on extreme-event accuracy (s2) and detection (s3).

Prior work has established that outage counts are over-dispersed and well-modeled by negative binomial regression [1], with zero-inflated extensions for the point mass at zero [2, 3]. GAM extensions offer flexible covariate effects [4] at the cost of unbounded predictions, while gradient-boosted trees [5] have demonstrated strong performance on high-dimensional meteorological inputs. Our contribution is a routing-based ensemble that partitions the outage space into four data-driven regimes, trains specialized models for each, and combines them with a meta-learner that corrects routing errors at test time using out-of-fold disagreement signals.

2 Data and Exploratory Analysis

The dataset provides hourly outage counts and 109 meteorological variables for all 83 Michigan counties from April 1 to June 30, 2023 (2,161 hourly timestamps per county), with no missing values in any field. Weather variables span near-surface temperature, humidity, pressure, wind components and gusts, convective instability indicators (CAPE, lifted index, CIN), precipitation, cloud diagnostics, and land-surface characteristics including vegetation type and leaf area index.

Distributional characteristics. Table 1 summarizes the key statistics. The mean of 45.25 outages per county-hour against a standard deviation of 452.27 (coefficient of variation > 10) and a skewness of 26.37 confirm an extreme right tail, with a single county-hour reaching 23,346 outages. Among non-zero observations, the median is 15 against a mean of 152.8, confirming a bimodal character: a point mass at zero alongside a heavy-tailed count process.

Table 1: Summary statistics for hourly outage counts across all 83 counties and 2,161 hourly timestamps.

Statistic	Value	Statistic	Value
% Zero county-hours	70.47%	95th percentile	100
Mean	45.25	99th percentile	933
Median	0	Maximum	23,346
Std. deviation	452.27	Skewness	26.37

Event-driven temporal structure. No consistent hour-of-day or day-of-week cycle is present; the distribution is event-driven. The June 26, 2023 storm produced conditions far outside the background: CAPE of 212 J/kg (14.6$\times$ the seasonal mean), a near-zero lifted index, and PWAT of 34.85 mm (2.35$\times$ background). County-averaged outages peaked at 1,040 per hour and remained elevated across all 83 counties for approximately 24 hours.

Spatial heterogeneity. County-level mean outages span a 15-fold range (below 50 to nearly 800), reflecting heterogeneity in infrastructure density, grid topology, and vegetation cover. Spatial autocorrelation across 67 adjacent county pairs yields a mean Pearson correlation of 0.175 (standard deviation 0.274), with most pairs weakly correlated—motivating within-county lag features over a spatially explicit model.

3 Methodology

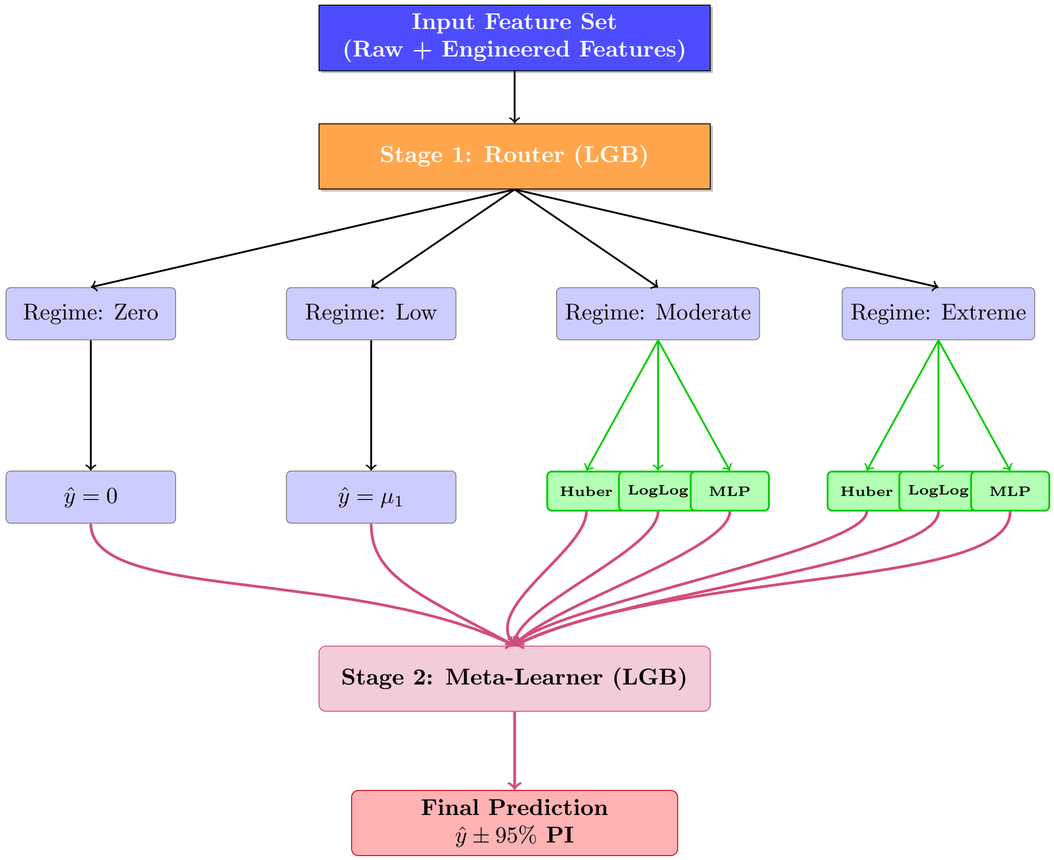
3.1 Problem Formulation and Routing Architecture

The full conditional distribution is approximated as a mixture over four magnitude regimes, as illustrated in the system architecture shown in Figure 1. Let $y_{c,t} \in \mathbb{Z}_{\geq 0}$ denote the hourly outage count for county $c \in \{1, \dots, 83\}$ at forecast hour $t \in \{1, \dots, 48\}$. The full conditional distribution is approximated as follows:

$$P(y_{c,t} \mid \mathbf{x}_{c,t}) = \sum_{k=0}^{3} P(y_{c,t} \mid \text{class}_k, \mathbf{x}_{c,t}) \cdot P(\text{class}_k \mid \mathbf{x}_{c,t}). \tag{1}$$

Class 0 covers zero outages; Class 1 covers low activity $[1, 10)$; Class 2 covers moderate outages $[10, P_{80}]$; and Class 3 covers the extreme tail $y > P_{80}$. The Class 0 boundary follows naturally from the structural zero-inflation. The three non-zero regimes were determined by k-means clustering on

Figure 1: Architecture of the Four-Stage Routing Ensemble.



the non-zero outage distribution, selecting $k = 3$ via silhouette score maximization and validating boundaries through inter-cluster Wasserstein distance and feature divergence. The threshold $P_{80} \approx 299$ is re-computed within each fold on training data only, preventing information leakage

Direct Multi-Step Forecasting. All components of the routing architecture—the classifier and the class-specific regressors—are trained in a direct multi-step configuration. Specifically, 48 separate models are fit for each component, one for each forecast horizon $h \in \{1, \ldots, 48\}$. The h-ahead model is trained to predict $y_{c,t+h}$ using the feature vector available at time t, ensuring that no future weather observations enter the input. At test time, the final feature vector from the end of the training window is provided to all 48 horizon-specific models, each producing a single prediction for its respective hour. The meta-learner then combines these horizon-specific outputs using the disagreement and soft-routing signals learned during training. This design avoids autoregressive rollouts and eliminates error accumulation across the 48-hour horizon.

3.2 Feature Construction

The base feature set comprises the 109 raw meteorological variables and the tracked household count. Gradient-boosted trees discover non-linear interactions through split-based gain without handcrafted transformations. Temporal lag features:

$$\mathbf{x}_{\text{lag}} = \left[y_{c,t-h}\right]_{h \in \{1,2,3,4,5,6,7,10,24,48\}}, \tag{2}$$

capture outage persistence across multi-hour storm sequences. Calendar features (hour, day of week, minute of day), county identifier, and exponentially-weighted moving average (EWM) transformations of all meteorological variables are also included, for a total input dimensionality of 349 features. All meteorological variables are shifted one hour backward relative to the target, ensuring no future observation enters the feature vector. This one-hour lagging aligns with the competition requirement that no future weather observations be used during the test window.

3.3 Stage 1: Multi-Class Router

A LightGBM classifier [5] estimates $\mathbf{p}_{c,t} = [P(\text{class}_k \mid \mathbf{x}_{c,t})]_{k=0}^{3}$ with routing by $\hat{c}_{c,t} = \arg\max_k p_k$. Class weights are set inversely proportional to class frequency for cost-sensitive learning. Class 0 predicts exactly zero; Class 1 predicts the within-class training mean. Regressor diversity is reserved for Classes 2 and 3, where preliminary experiments identified systematic tail underprediction as the primary failure mode of a single global model.

3.4 Stage 2: Regime-Specific Regressors

For Classes 2 and 3, three complementary regressors are trained. **LightGBM with Huber Loss** minimizes:

$$\mathcal{L}_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & |y - \hat{y}| \le \delta, \\ \delta\big(|y - \hat{y}| - \frac{\delta}{2}\big) & \text{otherwise,} \end{cases} \tag{3}$$

downweighting extreme residuals while preserving gradient signal for moderate values. **LightGBM with Log-Log Transformation** models $\tilde{y} = \log(1 + \log(1 + y))$, recovering predictions via $\hat{y} = \exp(\exp(\hat{\tilde{y}}) - 1) - 1$, yielding near-Gaussian residuals within Class 3 and stable gradients at extreme counts. **A Multilayer Perceptron** (hidden dimensions 128 and 64, ReLU activations, MSE loss, Adam with L2 decay) introduces smooth non-linear feature interactions beyond what tree splits approximate. The ensemble prediction is the arithmetic mean of successfully trained regressors for that class.

3.5 Stage 3: Bayesian Hyperparameter Optimization

Model parameters are tuned via the Tree-structured Parzen Estimator (TPE) in Optuna with successive halving pruning. The objective is F1 score on the most recent validation fold, representing conditions closest to the test window. Key parameters include tree depth, learning rate, regularization strength, and the Huber transition parameter δ, optimized independently for Classes 2 and 3.

3.6 Stage 4: Meta-Learner

A second LightGBM classifier is trained on the 10-dimensional out-of-fold meta-feature vector:

$$\mathbf{z}_{c,t} = \big[p_0,\, p_1,\, p_2,\, p_3,\; \hat{y}_L^{(1)},\, \hat{y}_L^{(2)},\, \hat{y}_L^{(3)},\; \hat{y}_H^{(1)},\, \hat{y}_H^{(2)},\, \hat{y}_H^{(3)}\big], \tag{4}$$

exposing router soft-probability uncertainty, regressor agreement or disagreement, and patterns in which model performs best for a given meteorological configuration. Training entirely on out-of-fold predictions prevents overfitting to in-sample behavior.

3.7 Validation Strategy and Uncertainty Quantification

Models are trained within a five-fold expanding window scheme, each fold adding approximately two weeks of data and holding out the following block. Fold boundaries mirror the day-of-week composition of the test window to ensure the Optuna objective is evaluated on temporally representative observations. Final test predictions average all five fold models.

For 95% prediction intervals, a fixed-width global residual quantile approach proved inadequate: prediction uncertainty scales with outage severity, and a single global offset generates intervals that are too narrow for extreme events and unnecessarily wide for quiet periods. We therefore adopt a magnitude-scaled construction:

$$L_{c,t} = \max\bigl(0,\ 0.75\,\hat{y}_{c,t}\bigr), \qquad U_{c,t} = \max\bigl(L_{c,t} + 3,\ 1.35\,\hat{y}_{c,t}\bigr). \tag{5}$$

The lower bound at 75% and upper bound at 135% of the point forecast reflect the empirical heteroscedasticity of outage prediction errors, where residual variance grows with outage magnitude. The minimum width of 3 outages prevents degenerate near-zero intervals, which would incur a 40$\times$ Winkler penalty at $\alpha = 0.05$ for any missed actual event. The scale parameters (0.75, 1.35) and minimum width (3) were jointly tuned using out-of-fold residuals from all five validation folds, targeting approximately 95% empirical coverage across the 175,296 validation samples while minimizing expected Winkler score given the competition's penalty structure.

4 Feature Contribution and Model Interpretability

4.1 Methodology

Feature importance is quantified using gain-based (Gini) importance from the native LightGBM API, measuring the average impurity reduction when a feature is used as a split variable. Scores are aggregated across all 336 trained models (7 model types $\times$ 48 hourly variants), covering 175,296 training samples. The feature set comprises 349 variables: 109 raw meteorological variables, 219 EWM transformations, 10 temporal outage lags, and 11 calendar and identifier features. Importances are compared separately for the routing classifier, moderate-severity regressors (Class 2, LOW), and extreme-severity regressors (Class 3, HIGH). The LOW$\rightarrow$HIGH importance shift provides a principled test for causal weather influence: a feature with substantially larger importance in HIGH models differentiates *how severe* an event becomes, not merely whether one is occurring.

4.2 Detection versus Severity: A Structural Feature Separation

The central interpretability finding is a complete structural separation between features driving outage detection and those driving outage severity. Table 2 shows the top features for the routing classifier and the extreme-severity regressor (Huber HIGH). The two feature sets share no variables.

The classifier answers "*when* is an outage occurring?" through temporal autocorrelation: time elapsed since the last non-zero outage and the recent lagged counts together account for the majority of its importance. This is not circular reasoning—storms persist for hours, so recent outage activity is a physically valid signal that the system remains active. The severity regressors answer "*how severe* will it be?" entirely through meteorological physics, with zero temporal features among their top drivers. This separation validates the routing architecture: detection and severity are genuinely different sub-problems requiring different feature signals, and no single model can optimize both.

Table 2: Top features for the routing classifier versus the extreme-severity regressor (Huber HIGH). Zero overlap between detection and severity drivers. Scores are gain-based Gini values.

Feature	Role	Classifier	Huber HIGH
`time_since_nonzero`	Temporal state	0.131	–
`outage_lag_1h`	Temporal autocorr.	0.115	–
`outage_lag_2h`	Temporal autocorr.	0.032	–
`lftx4_ewm_mean`	Conv. instability	–	0.033
`gh_4_ewm_mean`	Upper-level dyn.	–	0.020
`d2m_ewm_mean`	Dewpoint / moisture	–	0.014
`sh2_ewm_mean`	Near-surf. humid.	–	0.008

4.3 Top Three Severity-Driving Weather Features

Table 3 ranks the top features by LOW→HIGH importance shift across the Huber regressors, with LogLog Δ providing independent validation.

Table 3: Top three weather features by LOW→HIGH importance shift. A large positive Δ identifies features that drive extreme-event *severity*. LogLog Δ is an independent cross-framework validation.

Feature	LOW	HIGH	Huber Δ	LogLog Δ
`lftx4_ewm_mean`	0.00564	0.03325	+0.02761	+0.02764
`gh_4_ewm_mean`	0.00276	0.01983	+0.01707	+0.00985
`sh2_ewm_mean`	0.00256	0.00823	+0.00567	+0.03538

1. Lifted Index (lftx4_ewm_mean). The lifted index (LI) measures the temperature difference between an adiabatically lifted parcel and the ambient environment at 400 mb; negative values confirm atmospheric instability and potential for deep convection. The EWM-smoothed variant captures accumulated thermodynamic energy in the hours preceding storm onset, rather than an instantaneous reading. LI is a direct operational proxy for CAPE, and its Huber Δ of +0.028—the largest of any feature, consistent across both the Huber and LogLog frameworks—confirms it as the primary thermodynamic driver of severe outages in Michigan. During the June 26, 2023 event, the lifted index fell to near zero alongside the extreme CAPE of 212 J/kg.

2. Geopotential Height at 400 mb (gh_4_ewm_mean). Anomalously low geopotential heights at 400 mb indicate an upper-level trough, which supports surface cyclogenesis, enhances wind shear, and organizes convection into linear squall lines most responsible for widespread infrastructure damage. Unlike surface variables, 400 mb height captures the large-scale steering flow that governs storm track and duration—directly controlling how many counties are affected. Its Huber Δ of +0.017, confirmed by a LogLog Δ of +0.010, establishes it as the dominant large-scale dynamical predictor of extreme outage magnitude.

3. Specific Humidity at 2 m (sh2_ewm_mean). Near-surface specific humidity measures the low-level moisture supply fueling convective precipitation intensity. High values sustain heavier precipitation, which saturates soils, reduces tree root anchorage, and extends outage risk via secondary vegetation-damage pathways beyond those attributable to wind alone. The feature's LogLog Δ of +0.035 is the largest of any feature in that framework, indicating that near-surface moisture is especially important at the most extreme tail where precipitation-driven secondary damage is most pronounced.

4.4 Normal-Regime versus Extreme-Regime Differences

Under normal conditions (below τ_c), broad atmospheric state variables dominate the classifier's weather-signal component: mean sea-level pressure (`mslma`), leaf area index (`lai`), and vegetation cover (`veg`) reflect background infrastructure vulnerability from routine vegetation-contact outages. During extreme events, the lifted index, geopotential height, and near-surface humidity completely displace these predictors, reflecting that severe convective systems impose acute thermodynamic and mechanical damage overwhelming any persistent vegetation effect.

4.5 Limitations

Gain-based importance can overweight features with many unique values relative to binary or categorical predictors, and permutation importance—while more theoretically sound—was computationally intractable across 336 models. EWM importance is attributed jointly to the meteorological variable and its temporal accumulation window, so individual feature attribution is distributed across raw and EWM variants. The training window is dominated by a single storm event, limiting the diversity of atmospheric configurations in the extreme regime. Finally, the lifted index, geopotential height, and specific humidity are mutually correlated within convective environments; their importance split reflects a decomposition of a jointly important thermodynamic cluster rather than three fully independent physical signals.

5 Results

5.1 Validation Performance

Table 4 reports out-of-fold performance for the base Stage 2 ensemble (without meta-learner) and the full four-stage model, across all four competition metrics. The meta-learner produces substantial improvements on every metric.

Table 4: Out-of-fold validation performance. Lower is better for s1, s2, s4; higher is better for s3. % Improvement reflects the meta-learner's lift over the base ensemble.

Model configuration	s1 (RMSE)↓	s2 (Tail RMSE)↓	s3 (F1)↑	s4 (Winkler)↓
Stage 2 base ensemble	3.971	43.497	0.675	89.378
Full model (all 4 stages)	2.256	27.793	0.696	9.495
Improvement	43.2%	36.1%	3.2%	89.4%

The meta-learner's 43.2% reduction in normal-case RMSE (s1) and 36.1% reduction in tail RMSE (s2) demonstrate that exploiting regressor disagreement and soft routing probabilities substantially corrects for routing errors at the Class 2/3 boundary. The Winkler score improvement of 89.4%—from 89.38 to 9.49—is the most striking result and reflects the magnitude-scaled interval design: the global residual quantile approach assigns the same fixed offset to all predictions regardless of their magnitude, generating intervals that are far too wide for small predictions (incurring proportional Winkler width penalties) while being too narrow for extreme ones. The adaptive scaling directly addresses this heteroscedasticity.

For binary outage detection (s3), the full model achieves an F1 of 0.696, with precision of 0.893 and recall of 0.571. The high precision indicates that predicted non-zero events are reliable; the lower recall reflects that a meaningful fraction of actual outage county-hours are routed to Class 0 by the classifier, particularly for low-signal events where recent outage history is absent and meteorological conditions are ambiguous. The imbalance between precision and recall is a direct consequence of the asymmetric class distribution and represents the principal remaining limitation of the current routing design.

5.2 Test Predictions

Final test predictions cover the 48-hour window from June 29, 01:00 to July 2, 00:00. All predictions are clipped to $[0, \infty)$ and 95% prediction intervals are applied using the magnitude-scaled calibration fitted on the full set of out-of-fold residuals.

6 Conclusion

This paper has presented a routing-based ensemble for county-level power outage forecasting built around the structural properties of the target distribution: severe zero-inflation, heavy tails, and a data-generating process dominated by rare convective events. The four-stage framework—data-driven regime identification, specialized regressors per regime, Bayesian hyperparameter optimization, and a meta-learner trained on out-of-fold predictions—avoids the gradient dilution that prevents any single model from excelling on both normal-period and extreme-event accuracy. Magnitude-scaled adaptive prediction intervals, whose parameters are tuned to minimize the Winkler penalty while maintaining 95% empirical coverage, reduce the Winkler score by 89% relative to a global residual quantile baseline.

Feature importance analysis reveals a physically coherent finding that directly validates the architecture: detection and severity prediction rely on entirely different feature sets. The routing classifier uses temporal autocorrelation to identify *when* extremes occur, while severity regressors rely on meteorological physics—the Lifted Index, 400 mb geopotential height, and near-surface specific humidity—to determine *how severe* an event becomes. This zero-overlap structure confirms that the two tasks cannot be collapsed into a single model without sacrificing performance on at least one.

The results point to several concrete directions for improvement. The recall of 0.571 against a precision of 0.893 indicates that roughly 43% of outage events are missed by the routing classifier, predominantly low-signal events where recent outage history is absent and the meteorological signal is weak. Threshold calibration and explicit cost-sensitive weighting on the Class 0 boundary could recover a meaningful fraction of these missed detections without substantially degrading precision. The maximum absolute validation error of 1,412 outages confirms that the most extreme individual county-hours remain difficult to capture; applying conformal prediction or extreme value theory to Class 3 alone could produce better-calibrated intervals for this regime without widening intervals across the majority of quiet county-hours. Finally, the magnitude-scaled prediction intervals currently use globally fitted scale parameters; fitting separate parameters for each severity class would account for the distinct residual distributions of the moderate and extreme regimes and likely yield further Winkler score reductions.

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